



इलेक्ट्रॉनिक्स एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY

Advanced Certificate Course (ACC) in HPC

NCVET Certified

Overview

High Performance Computing (HPC) plays an important role in both scientific advancement and economic competitiveness of a nation - making production of scientific and industrial solutions faster, less expensive, and of higher quality. HPC is a key component in many applications including designing of vehicles, aeroplanes, discovery of drugs, discovery and extraction of new energy sources like oil and natural gas, weather forecasting, and many more

C-DAC is the pioneer in the area of the supercomputing and has established HPC infrastructure across the nation in the past 35+ years through its PARAM supercomputers. C-DAC has been involved in creating HPC skilled manpower who contribute to nation building by solving computationally challenging problems in several areas of practical significance.

C-DAC, under the aegis of Ministry of Electronics & Information Technology (MeitY), Govt. of India has announced an Advanced Certificate Course (ACC) to develop skilled manpower in the area of HPC-AI. This course is developed as part of the Centre for HPC Upskilling & Knowledge-sharing (C-HUK) activity under the Human Resource Development (HRD) initiatives of the National Supercomputing Mission (NSM).

The course provides comprehensive training in Linux, HPC architecture, cloud computing, and advanced programming techniques. It emphasizes parallel programming using C++, Python, and modern accelerator technologies such as CUDA, OpenCL, and OpenACC. Learners will also gain hands-on experience in machine learning and deep learning model deployment, statistical analysis, and data visualization using Power BI, preparing them for High Performance, Data-intensive computing environments.

Salient Features of ACC

- 1 The course is offered to candidates belonging to **SC, ST, Women, EWS, General Category**
- 2 No registration or training fee is charged for any candidates
- 3 A monthly stipend of **₹10,000/-** will be provided to **SC, ST, Women and EWS candidates only**
- 4 Evaluation and Assessment will be done by C-DAC
- 5 Blockchain based Certificate will be issued on successful completion of the course
- 6 Access to PARAM Supercomputing facility for Hands-on during the course

Major topics covered:

Operating system, Shell Scripting

Computer Networks

HPC and its Architecture

Tools (Profilers, Debugger, Monitoring)

Analytics & Statistics using Python

Introduction to ML, DL, and OpenVino

HPC and Parallel Programming

OpenMP, MPI libraries

CUDA, OpenCL, OpenACC